

08 Geometric Transformations



8.7 Affine Transformation

Homogeneous coordinates unify translation, scaling, rotation, and shearing; 3×3 matrices for planar affine transforms, 4×4 matrices for 3D affine transforms

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Distinguish linear transformations from affine transformations: the former fix the origin, the latter add translation

2

Introduce homogeneous coordinates: unify translation and every earlier geometric operation as matrix multiplication

3

Use 3×3 matrix multiplication to complete planar affine transformations

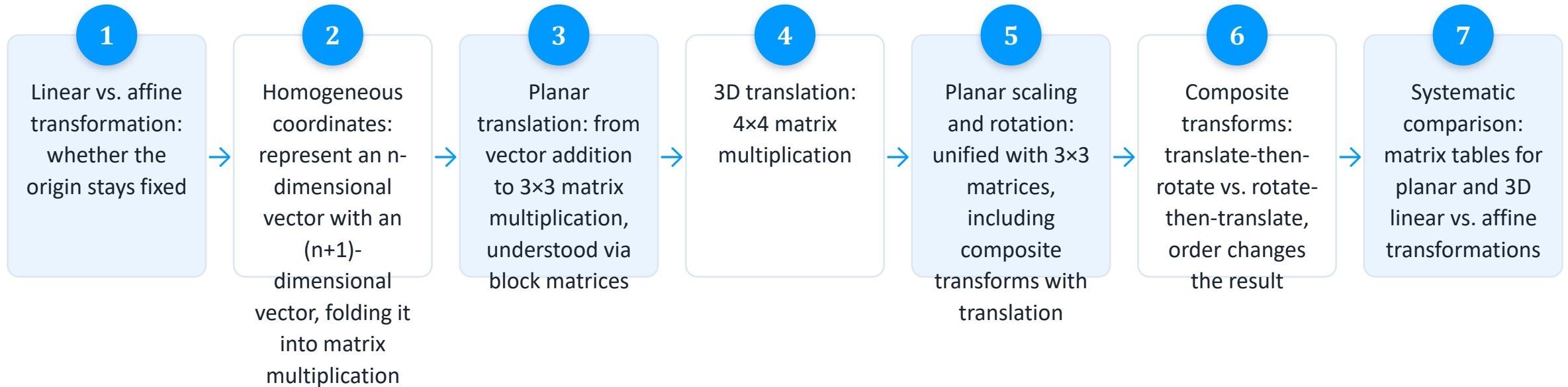
4

Use 4×4 matrix multiplication to complete 3D affine transformations

5

Understand affine transformation matrix multiplication via block matrices

From Homogeneous Coordinates to Planar and 3D Affine Transforms



Seven Angles for Reading Every Transform in This Chapter

- ① Geometric view of matrix multiplication: how a matrix transforms the zero vector $\mathbf{0}$, unit vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and the all-ones vector $\mathbf{1}$
- ② Check the determinant: is its absolute value greater than 1; is it positive, negative, or zero; determinants of diagonal/triangular matrices
- ③ Read area or volume change from the determinant
- ④ Check invertibility via the inverse matrix; inverses of special matrices
- ⑤ View parallelogram grids and parallelepiped shape changes from a linear-combination angle; does dimension reduction occur
- ⑥ Check from a basis viewpoint whether the transform matrix forms a basis, and what kind of basis
- ⑦ Properties of the transform matrix: symmetric, diagonal, orthogonal, invertible; does matrix multiplication commute

Affine Transformation Adds Translation on Top of a Linear Transformation

$$\mathbf{z} = \mathbf{Ax} \quad \text{vs} \quad \mathbf{z} = \mathbf{Ax} + \mathbf{t}$$

- A linear transformation can only scale, rotate, or shear — the origin never moves: it always maps to the origin
- An affine transformation can add translation on top of a linear transformation, moving the origin to any other location

Written as Column Vectors, Translation Is Simply Vector Addition

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \mathbf{t} \quad \mathbf{t} = \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

- $\mathbf{t}$ is the translation vector: t_1 is the horizontal shift, t_2 is the vertical shift
- Translation does not change the shape or size of a figure — only its position

Figure 1 · Vector Addition Completes Planar Translation

- The square shifts as a whole along vector $\mathbf{t}$; shape and size stay unchanged

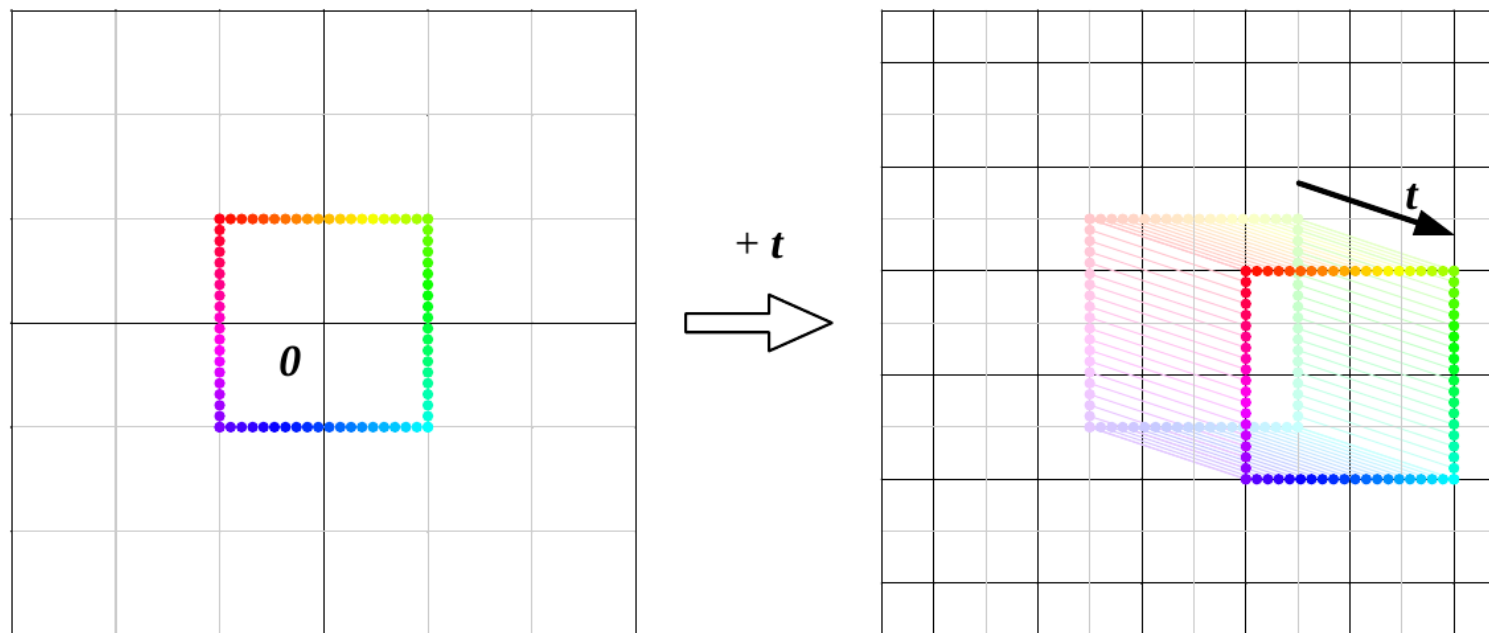


图 1. 向量加法完成平面平移操作

Homogeneous Coordinates Fold Translation into Matrix Multiplication

$$[z_1; z_2; 1] = A_{3 \times 3} @ [x_1; x_2; 1]$$

- Homogeneous coordinates represent an originally n-dimensional vector with an (n+1)-dimensional vector
- A 2×2 matrix alone cannot describe translation; a 3×3 matrix can uniformly express translation, scaling, rotation, and their composition

Using a 3×3 Matrix, Planar Translation Is the Following Matrix Product

$$[z_1; z_2; 1] = [1, 0, t_1; 0, 1, t_2; 0, 0, 1] @ [x_1; x_2; 1]$$

- t_1 is the horizontal shift of the figure, t_2 is the vertical shift
- The inverse of translation is also a matrix product: an upper-triangular matrix is invertible iff its diagonal entries are nonzero, and its inverse stays upper-triangular

Figure 2 · 3×3 Matrix Multiplication for Planar Translation and Its Inverse

- Translate along t , then along $-t$; the figure returns to its original position

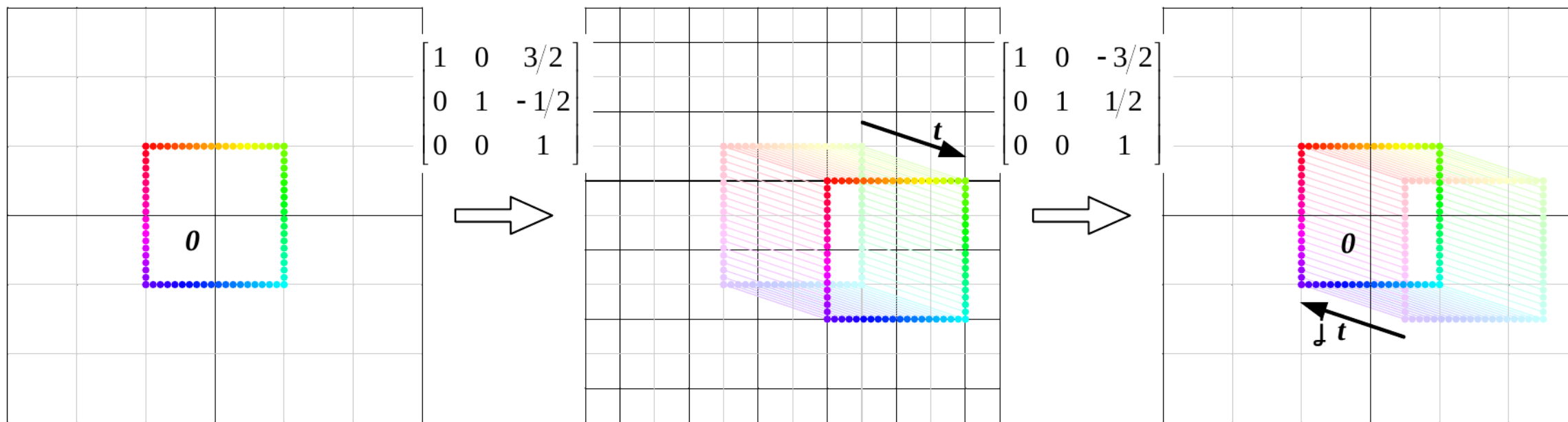


图 2. 利用 3×3 矩阵乘法完成平面平移操作、逆操作

Block Matrix Multiplication Makes the Translation Matrix Easy to Understand

$$[I_{2 \times 2}, \mathbf{t}; \mathbf{0}_{1 \times 2}, 1] @ [\mathbf{x}; 1] = [I_{2 \times 2}\mathbf{x} + \mathbf{t}; 1] = [\mathbf{z}; 1]$$

- Split the 3×3 translation matrix into the identity block I , translation vector $\mathbf{t}$, and zero row $\mathbf{0}$
- This directly reveals the addition equation $\mathbf{x} + \mathbf{t} = \mathbf{z}$ — block matrix multiplication restores the essence of vector addition

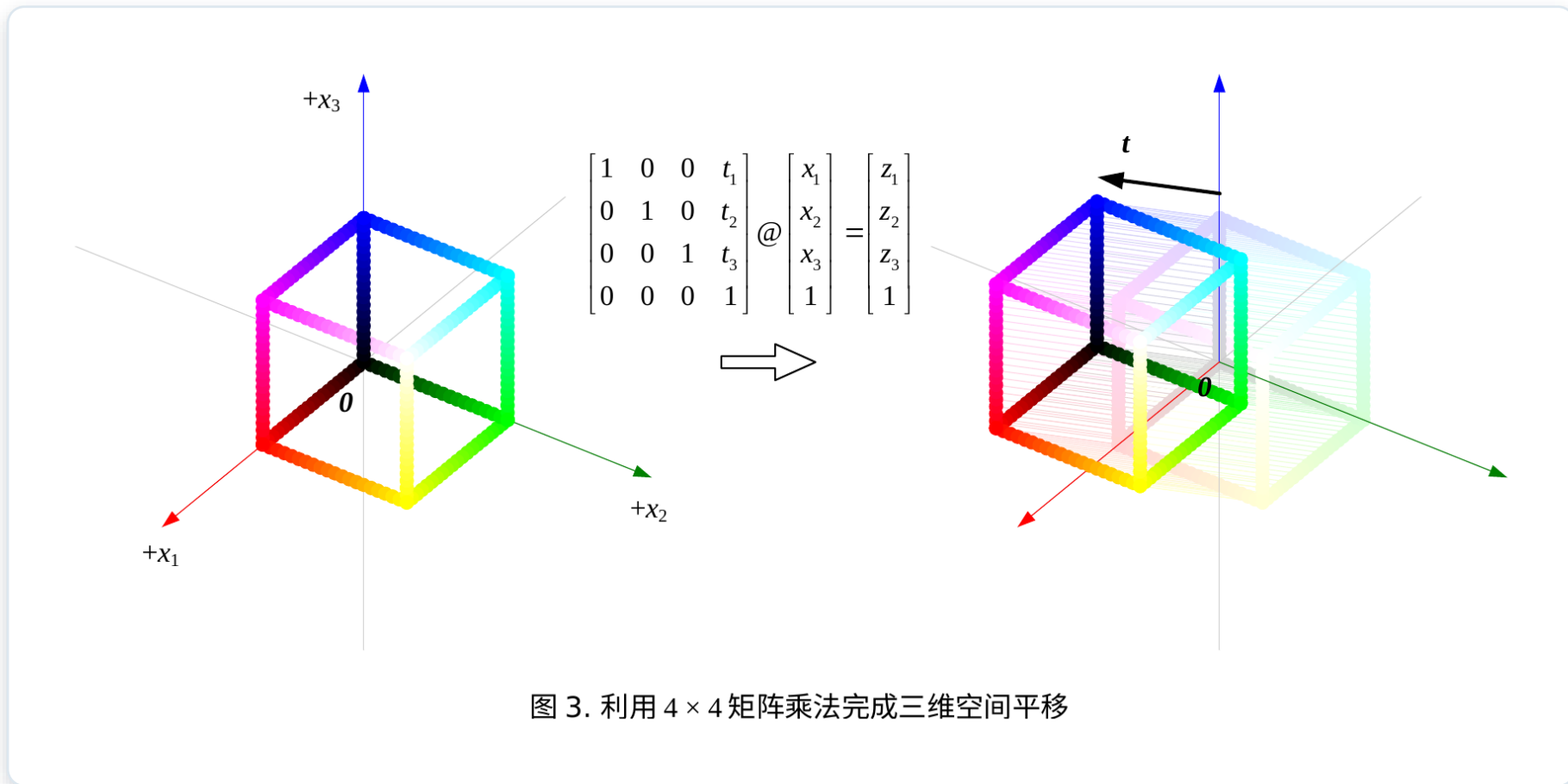
Likewise, 3D Geometric Transforms Can Use 4×4 Matrix Multiplication

$$[z_1; z_2; z_3; 1] = [1, 0, 0, t_1; 0, 1, 0, t_2; 0, 0, 1, t_3; 0, 0, 0, 1] @ [x_1; x_2; x_3; 1]$$

- 3D homogeneous coordinates represent an originally 3-dimensional vector with a 4-dimensional vector
- The inverse of translation is also a matrix product — simply negate t_1, t_2, t_3

Figure 3 · 4×4 Matrix Multiplication for 3D Translation

- The cube shifts as a whole along vector $\mathbf{t}$; shape and size stay unchanged



Using a 3×3 Matrix, Uniform Planar Scaling Is the Following Product

$$[z_1; z_2; 1] = [s, 0, 0; 0, s, 0; 0, 0, 1] @ [x_1; x_2; 1]$$

- $s > 1$ enlarges; $0 < s < 1$ shrinks; $s = 1$ leaves the figure unchanged
- When $s < 0$, scaling is combined with mirroring; the special case $s = 0$ is a projection that reduces dimension

Non-Uniform Scaling Uses Different Factors Along Different Directions

$$[z_1; z_2; 1] = [s_1, 0, 0; 0, s_2, 0; 0, 0, 1] @ [x_1; x_2; 1]$$

- s_1 is the horizontal scaling factor, s_2 is the vertical scaling factor
- The inverse of uniform scaling simply replaces each diagonal entry with $1/s$

Figure 4 · 3×3 Matrix Multiplication for Non-Uniform Scaling and Its Inverse

- Scale by different factors horizontally and vertically, then restore with the inverse scaling matrix

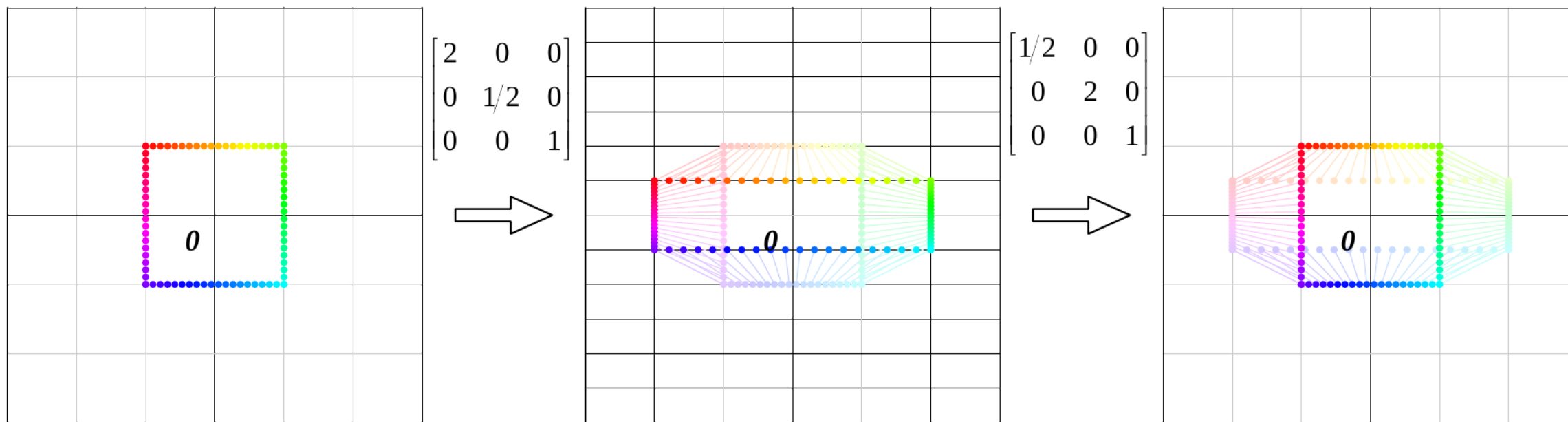


图 4. 利用 3×3 矩阵乘法完成非等比例缩放操作、逆操作

Using a 3×3 Matrix, Counterclockwise Rotation About the Origin Is the Following Product

$$[z_1; z_2; 1] = [\cos\theta, -\sin\theta, 0; \sin\theta, \cos\theta, 0; 0, 0, 1] @ [x_1; x_2; 1]$$

- ϑ is the rotation angle; counterclockwise is the positive direction
- The inverse of a rotation matrix simply replaces ϑ with $-\vartheta$ — rotation matrices are orthogonal

Figure 5 · 3×3 Matrix Multiplication for Rotation and Its Inverse

- Rotate 120° counterclockwise, then 120° clockwise; the figure returns to its original position

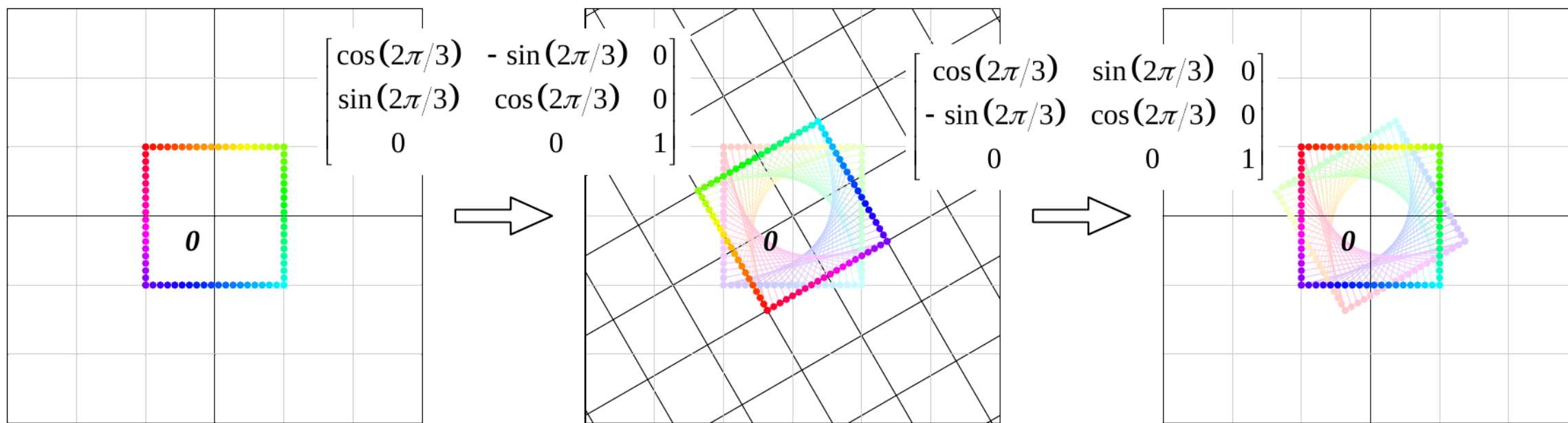


图 5. 利用 3×3 矩阵乘法完成旋转缩放操作、逆操作

Translate First, Then Rotate — the Translation Is Affected by the Rotation Angle

$$[\cos\theta, -\sin\theta, 0; \sin\theta, \cos\theta, 0; 0, 0, 1] @ [1, 0, t_1; 0, 1, t_2; 0, 0, 1]$$

- Matrix multiplication generally does not commute: a different order gives a different result
- Move the figure first, then rotate it about the origin; the effective translation ends up rotated along with the angle

Figure 6 · 3×3 Matrix Multiplication for Translate-Then-Rotate

- The square is first translated by $\mathbf{t}$, then rotated 120 degrees counterclockwise about the origin

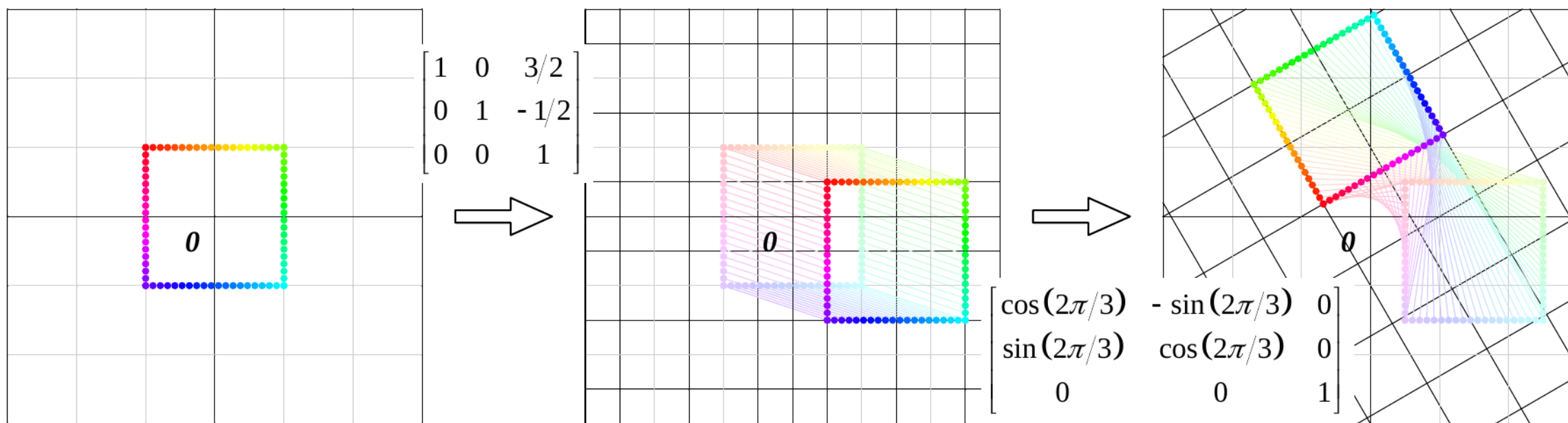


图 6. 利用 3×3 矩阵乘法完成先平移、再旋转

Rotate First, Then Translate — the Shift Follows the Global Axes, Unaffected by Rotation

$$[1, 0, t_1; 0, 1, t_2; 0, 0, 1] @ [\cos\theta, -\sin\theta, 0; \sin\theta, \cos\theta, 0; 0, 0, 1]$$

- Rotate about the origin first, then translate; the shift always follows the global coordinate axes, unaffected by the rotation
- This is exactly what makes affine transformations more flexible than linear ones: composition order decides the final effect

Figure 7 · 3×3 Matrix Multiplication for Rotate-Then-Translate

- The square is first rotated 120 degrees counterclockwise about the origin, then translated by t

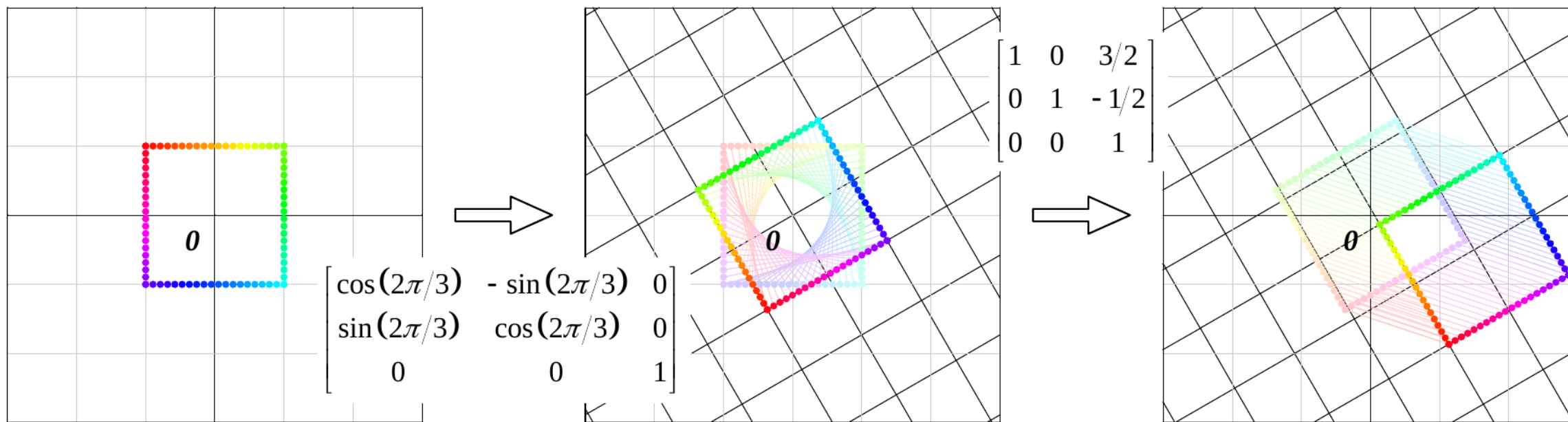


图 7. 利用 3×3 矩阵乘法完成先旋转、再平移

Table 1 (1/6) · No Transform, Translation

- Translation is not linear — the origin moves to a new location, which a 2×2 matrix cannot express

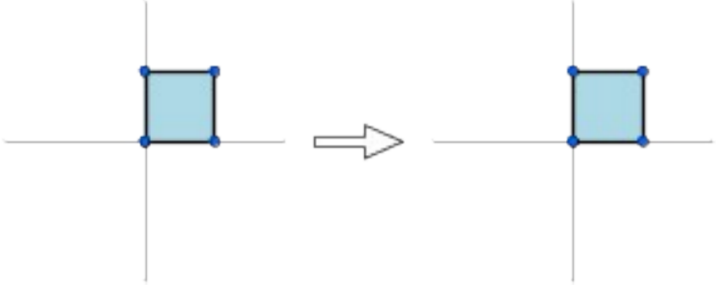
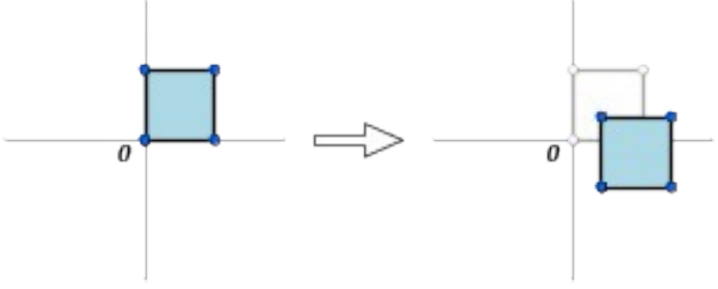
线性变换	仿射变换	图形变换
没变换 (单位矩阵) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
平移 不是线性变换	$\begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix}$	

Table 1 (2/6) · Uniform Scaling, Non-Uniform Scaling

- The 3×3 matrix pads a corner 1; its top-left 2×2 block matches the linear transform

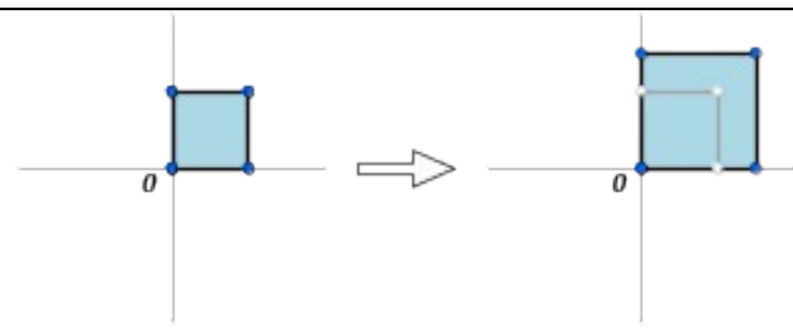
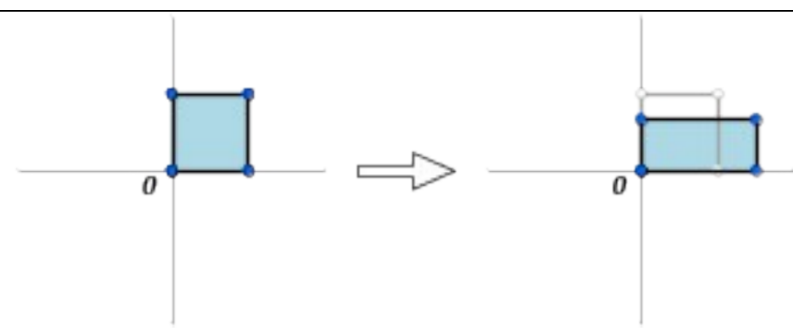
等比例缩放 $\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix}$ 行列式 s^2	$\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
非等比例缩放 $\begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$ 行列式 $s_1 s_2$	$\begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	

Table 1 (3/6) · Rotation, Shear Along the x1 Axis

- Rotation and shearing are both linear transforms with determinant 1 — figure area is unchanged

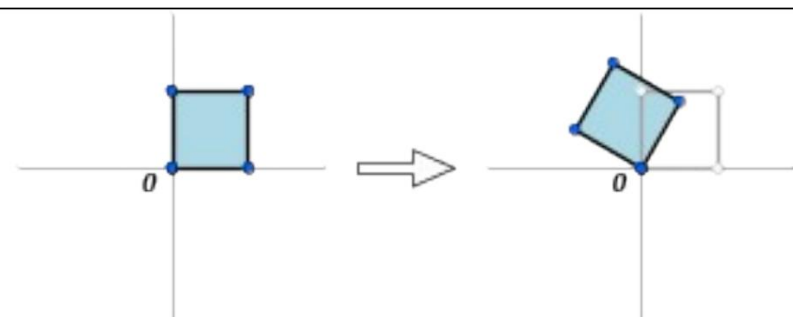
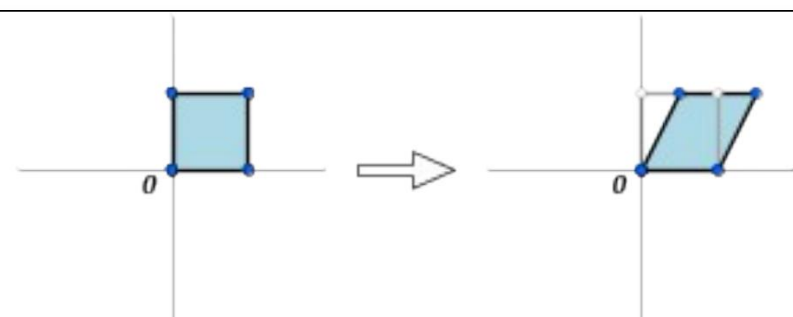
旋转 $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ 行列式 1	$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
沿 x_1 轴剪切 $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	

Table 1 (4/6) · Shear Along the x2 Axis, Orthogonal Projection

- Orthogonal projection has determinant 0 — an irreversible, dimension-reducing geometric operation

沿 x_2 轴剪切 $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} 1 & 0 & 0 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
正交投影 $\frac{1}{a_1^2 + a_2^2} \begin{bmatrix} a_2^2 & -a_1 a_2 \\ -a_1 a_2 & a_1^2 \end{bmatrix}$ (过原点) 直线法向量 $n = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ 行列式 0	$\begin{bmatrix} \frac{a_2^2}{a_1^2 + a_2^2} & \frac{-a_1 a_2}{a_1^2 + a_2^2} & 0 \\ \frac{-a_1 a_2}{a_1^2 + a_2^2} & \frac{a_1^2}{a_1^2 + a_2^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$	

Table 1 (5/6) · Projection Onto the x_1 Axis, Projection Onto the x_2 Axis

- Projecting onto an axis is likewise a determinant-0 reduction, compressing the figure onto a line

向 x_1 轴投影 $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ 行列式 0	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
向 x_2 轴投影 $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ 行列式 0	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	

Table 1 (6/6) · Mirroring

- Mirroring has determinant -1 : area is preserved, but orientation flips

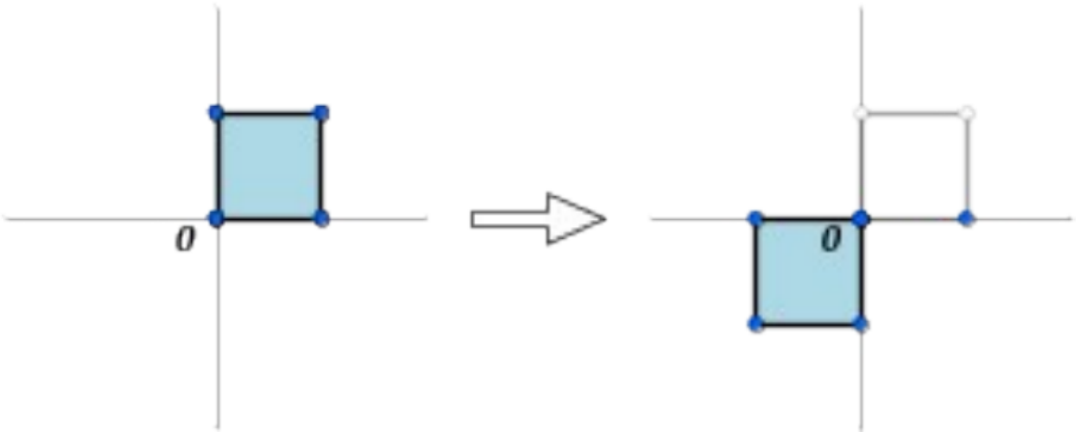
镜像 $\frac{1}{a_1^2 + a_2^2} \begin{bmatrix} a_2^2 - a_1^2 & -2a_1a_2 \\ -2a_1a_2 & a_1^2 - a_2^2 \end{bmatrix}$ (过原点) 直线法向量 $\mathbf{n} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ 行列式-1	$\begin{bmatrix} \frac{a_2^2 - a_1^2}{a_1^2 + a_2^2} & \frac{-2a_1a_2}{a_1^2 + a_2^2} & 0 \\ \frac{-2a_1a_2}{a_1^2 + a_2^2} & \frac{a_1^2 - a_2^2}{a_1^2 + a_2^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
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Table 2 (1/11) · No Transform, Translation

- 3D affine transforms use a 4x4 matrix: corner 1, top-left 3x3 block is the linear transform

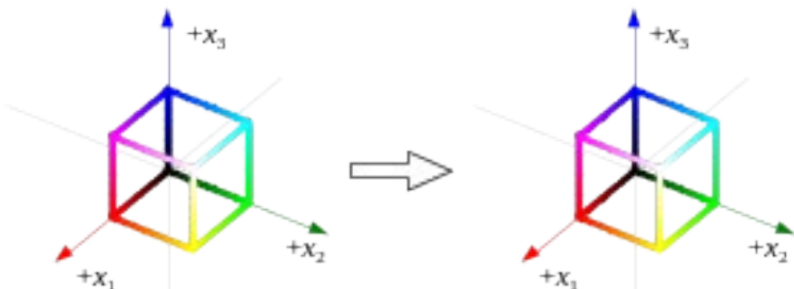
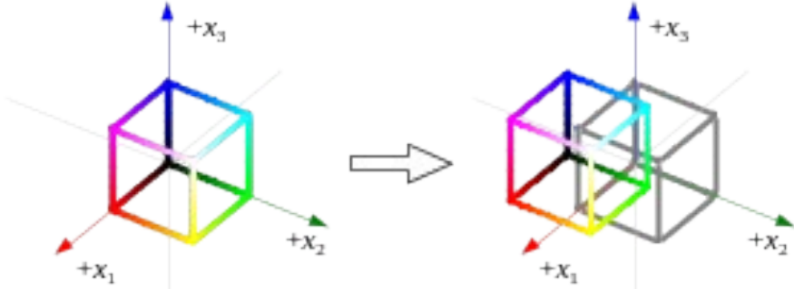
线性变换	仿射变换	示例
没变换 (单位矩阵) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
平移 不是线性变换	$\begin{bmatrix} 1 & 0 & 0 & t_1 \\ 0 & 1 & 0 & t_2 \\ 0 & 0 & 1 & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 2 (2/11) · Uniform Scaling, Non-Uniform Scaling

- 3D uniform scaling by s has determinant s^3 ; non-uniform scaling has determinant $s_1 \cdot s_2 \cdot s_3$

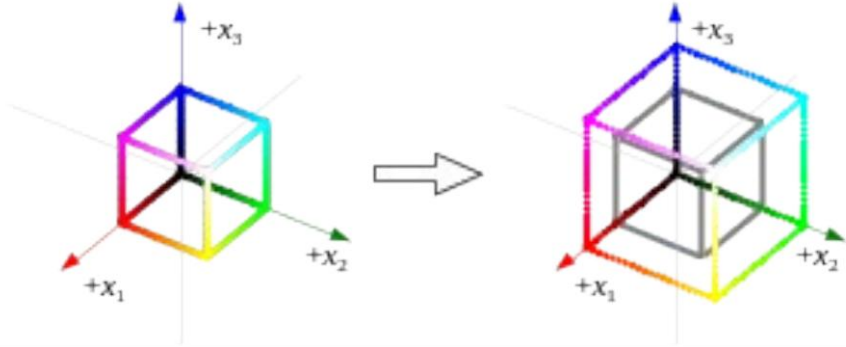
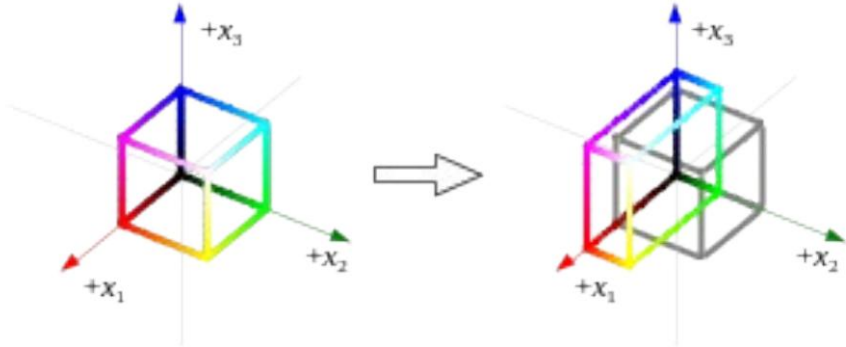
等比例缩放 s 倍 $\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix}$ 行列式 s^3	$\begin{bmatrix} s & 0 & 0 & 0 \\ 0 & s & 0 & 0 \\ 0 & 0 & s & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
非等比例缩放 $\begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & s_3 \end{bmatrix}$ 行列式 $s_1 s_2 s_3$	$\begin{bmatrix} s_1 & 0 & 0 & 0 \\ 0 & s_2 & 0 & 0 \\ 0 & 0 & s_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 2 (3/11) · Counterclockwise Rotation About the x1 Axis, About the x2 Axis

- 3D rotation matrices always have determinant 1; each axis places the 2×2 rotation block differently

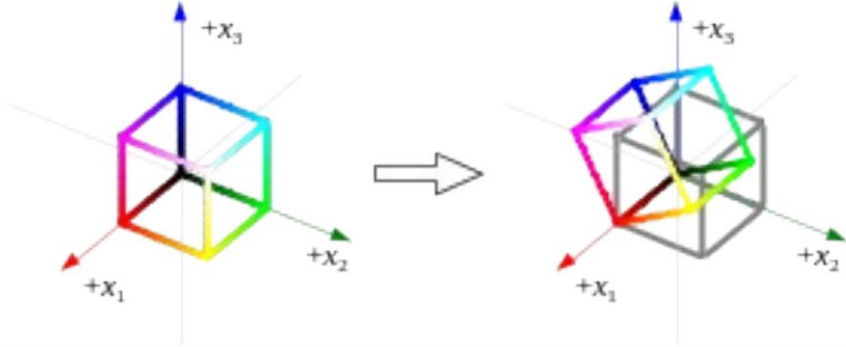
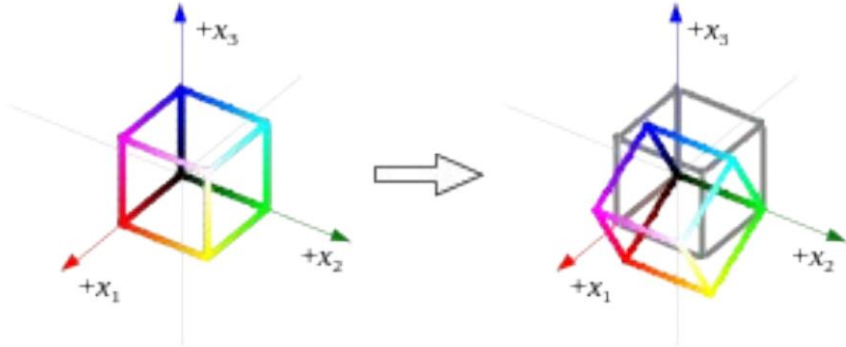
绕 x_1 轴逆时针旋转 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 \\ 0 & \sin \theta_1 & \cos \theta_1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_1 & -\sin \theta_1 & 0 \\ 0 & -\sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
绕 x_2 轴逆时针旋转 $\begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 \\ 0 & 1 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} \cos \theta_2 & 0 & \sin \theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 2 (4/11) · Counterclockwise Rotation About the x3 Axis, Mirroring About the x1x2 Plane

- 3D mirror matrices have determinant -1 , flipping the sign of the corresponding coordinate axis

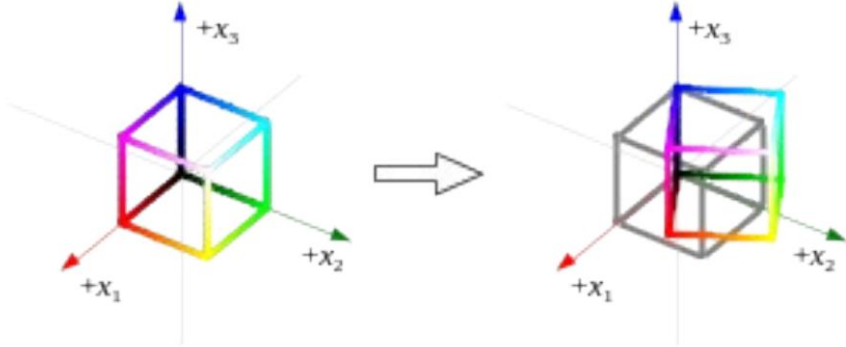
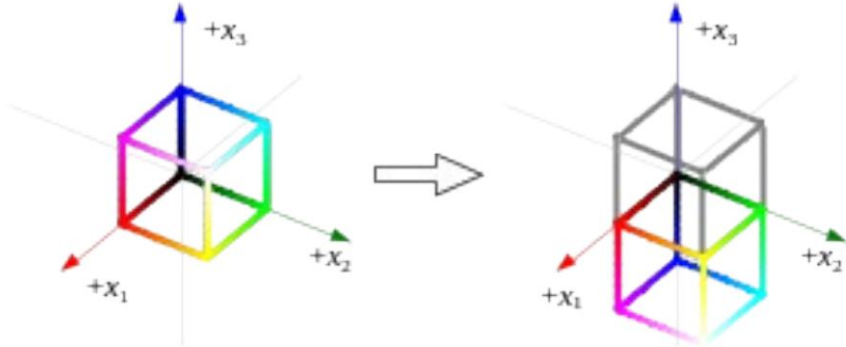
绕 x_3 轴逆时针旋转 $\begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 \\ \sin \theta_3 & \cos \theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 & 0 \\ \sin \theta_3 & \cos \theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
关于 x_1x_2 平面镜像对称 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ 行列式 -1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 2 (5/11) · Mirroring About the x_1x_3 Plane

- Mirroring about a coordinate plane flips only the single axis perpendicular to that plane

关于 x_1x_3 平面镜像对称

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

行列式-1

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

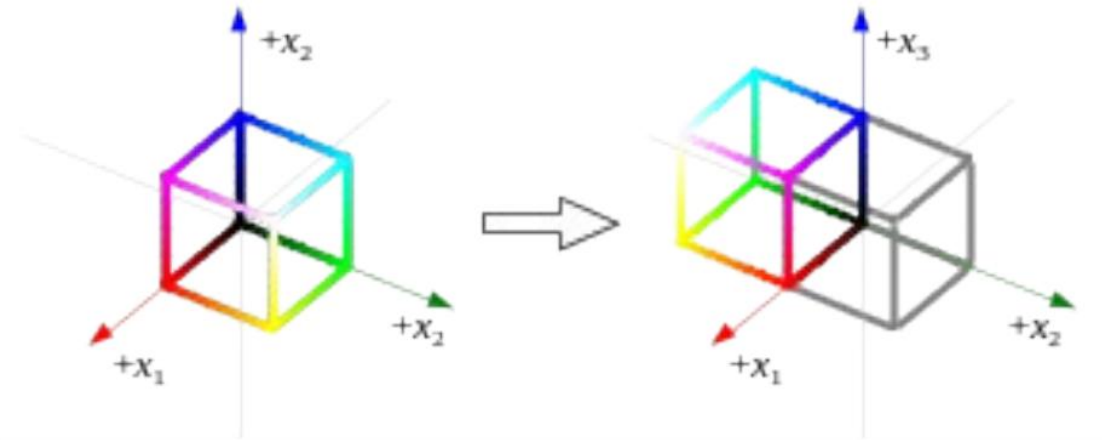


Table 2 (6/11) · Mirroring About the x_2x_3 Plane, Projection Onto the x_1x_2 Plane

- Projecting onto a coordinate plane has determinant 0 — an irreversible, dimension-reducing operation

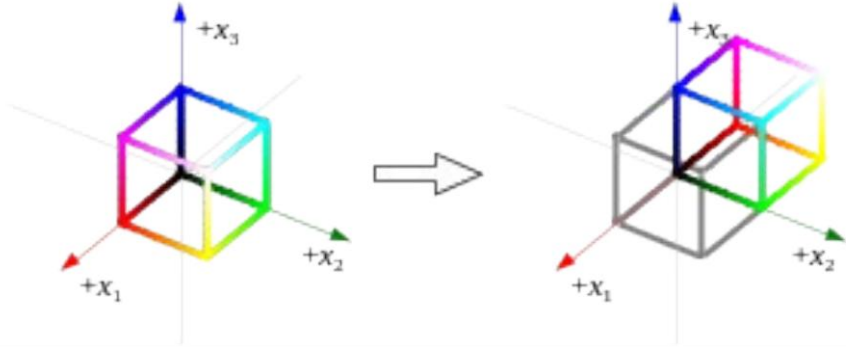
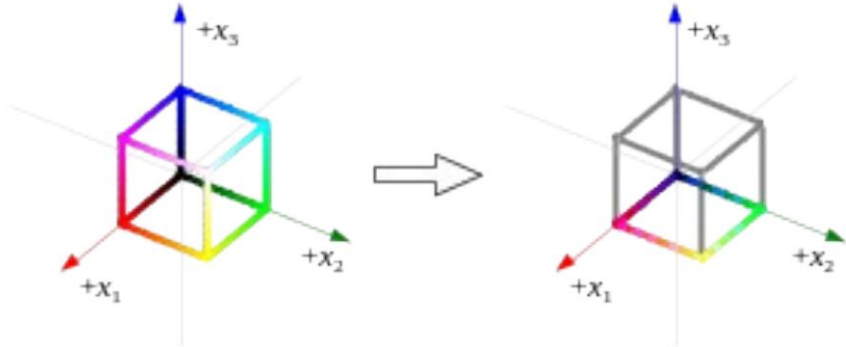
关于 x_2x_3 平面镜像对称 $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 行列式 -1	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
向 x_1x_2 平面投影 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ 行列式 0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 2 (7/11) · Projection Onto the x_1x_3 Plane, Projection Onto the x_2x_3 Plane

- A projection matrix keeps exactly two 1's on its diagonal, corresponding to the two preserved axes

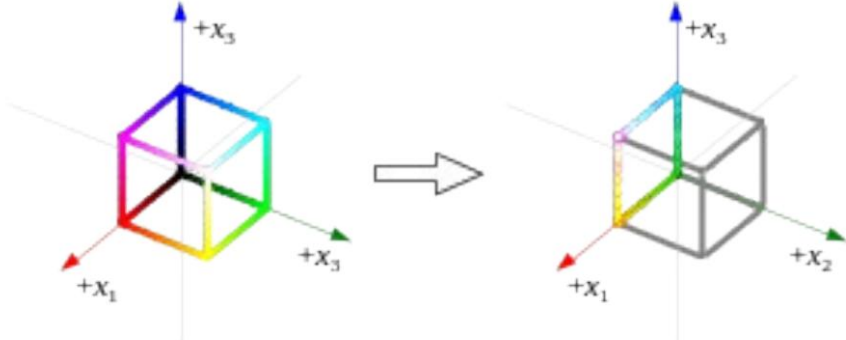
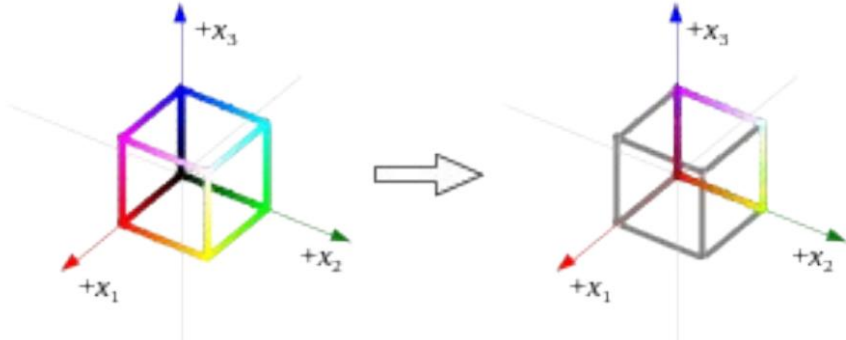
向 x_1x_3 平面投影 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 行列式 0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
向 x_2x_3 平面投影 $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 行列式 0	$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 2 (8/11) · Projection Onto the x_1 Axis, Projection Onto the x_2 Axis

- Projecting onto a coordinate axis compresses the 3D figure onto a line; determinant is 0

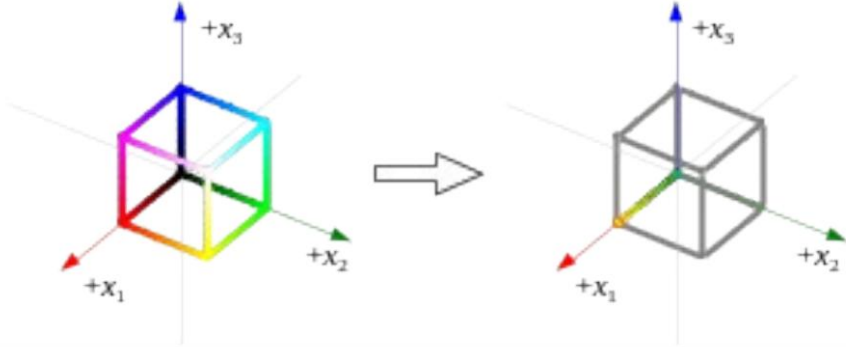
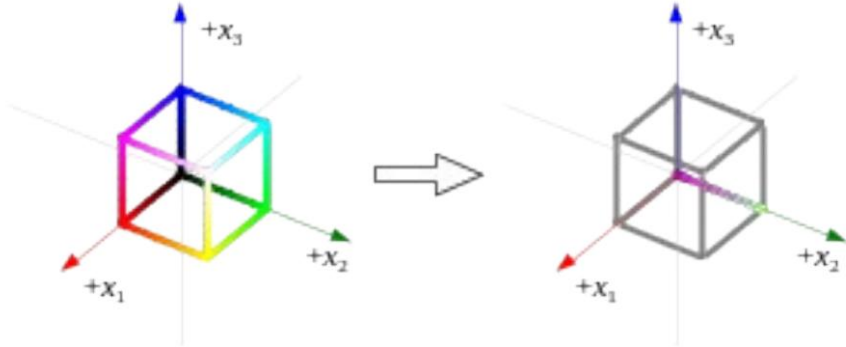
向 x_1 轴投影 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ 行列式 0	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
向 x_2 轴投影 $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ 行列式 0	$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 2 (9/11) · Projection Onto the x3 Axis

- Only the x3 coordinate is kept and the rest are zeroed — another irreversible reduction

向 x_3 轴投影

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

行列式 0

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

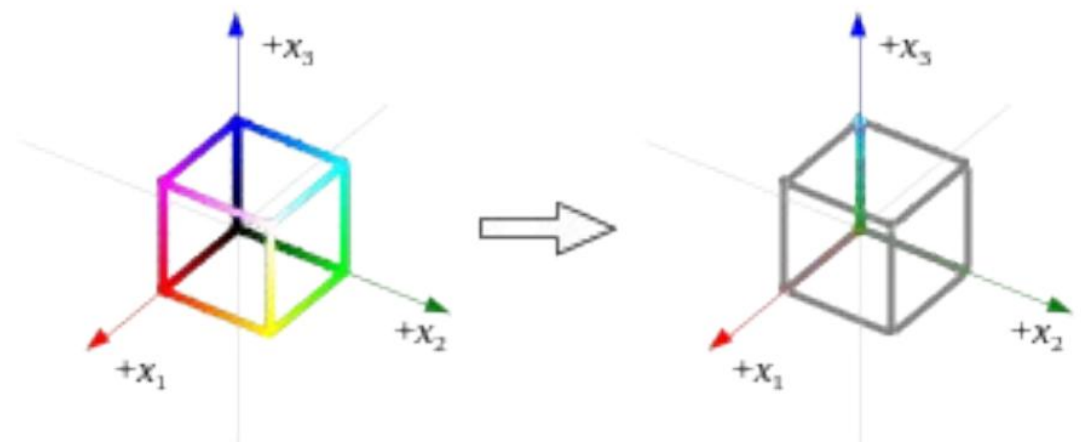


Table 2 (10/11) · Shear Along the x1 Axis, Shear Along the x2 Axis

- Each axis's shear has two reference directions; determinant is always 1, so volume is preserved

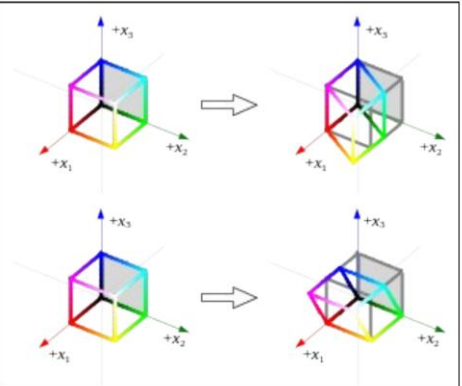
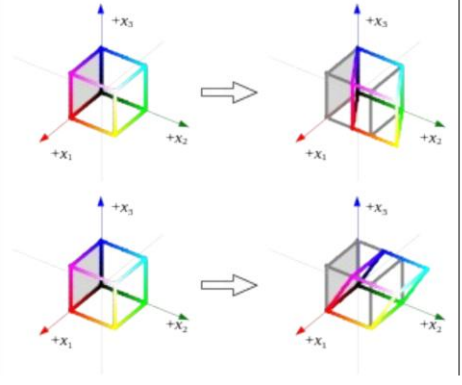
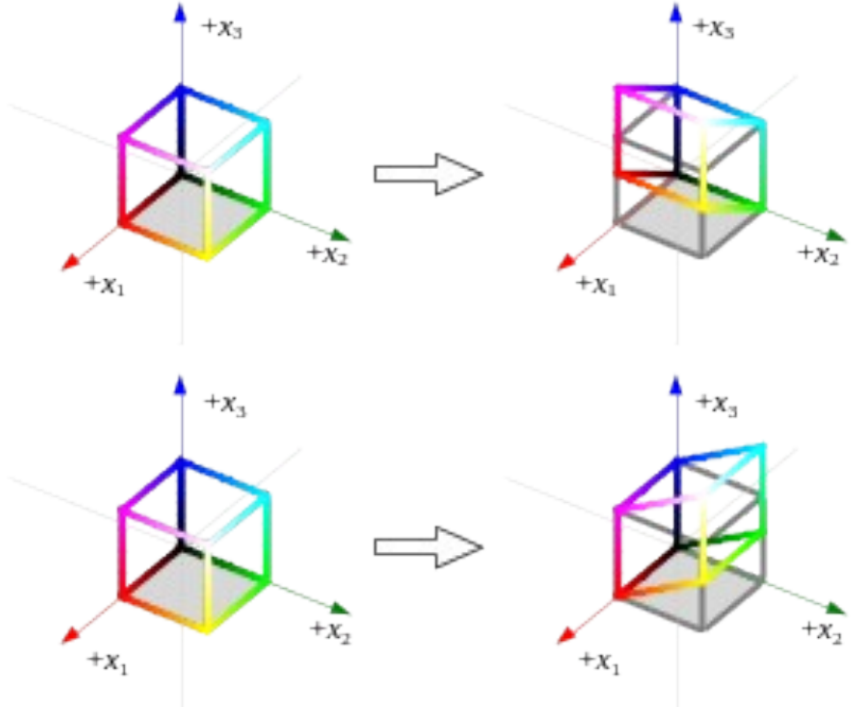
沿 x_1 轴剪切 $\begin{bmatrix} 1 & k_{1,2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & k_{1,3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} 1 & k_{1,2} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & k_{1,3} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
沿 x_2 轴剪切 $\begin{bmatrix} 1 & 0 & 0 \\ k_{2,1} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k_{2,3} \\ 0 & 0 & 1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ k_{2,1} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & k_{2,3} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 2 (11/11) · Shear Along the x3 Axis

- 3D shear closes this chapter, joining scaling, rotation, projection, and mirroring

沿 x_3 轴剪切 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ k_{3,1} & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & k_{3,2} & 1 \end{bmatrix}$ 行列式 1	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ k_{3,1} & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & k_{3,2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
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SUMMARY

What Does Matrix Multiplication Reveal About Affine Transformation?

Affine transform = linear transform + translation; a linear transform is the special case with a fixed origin

Homogeneous coordinates lift an n -dimensional vector to $n+1$ dimensions, letting translation become matrix multiplication too

Planar affine transforms use 3×3 matrices; 3D affine transforms use 4×4 matrices

Block matrix multiplication reveals the essence of the affine transform matrix: $\mathbf{Ax} + \mathbf{t} = \mathbf{z}$

Matrix multiplication generally does not commute: translate-then-rotate and rotate-then-translate give different results

Key Terms in This Section

Affine Transformation 仿射变换 —— a geometric transformation that adds translation on top of a linear transformation

Homogeneous Coordinates 齐次坐标 —— represents an n -D vector with an $(n+1)$ -D vector, folding it into matrix multiplication

Block Matrix 分块矩阵 —— dividing a matrix into sub-blocks to make a composite operation easier to understand

Rotation Angle 旋转角度 —— ϑ , counterclockwise is the positive direction

Commutative Law 交换律 —— generally does not hold for matrix multiplication; operation order affects the final result

Linear Transformation 线性变换 —— the special affine transformation that keeps the origin fixed

Translation Vector 平移向量 —— $\mathbf{t}$, controls the direction and distance a figure moves

Scaling Factor 缩放因子 —— s , controls how much a figure is enlarged or shrunk

Composite Transformation 组合变换 —— several geometric operations combined into one matrix via matrix multiplication

Dimension-Reducing Transform 降维变换 —— a projection-type transform with determinant 0

FURTHER STUDY

Suggested Further Reading

Review · 8.1 Geometric Transformations and Matrix Multiplication

This section's affine transforms build on every linear transform covered earlier in this chapter

Review · 8.5 Orthogonal Projection, 8.6 Reflection

Detailed coverage of the dimension-reducing and flipping transforms in Tables 1 and 2

Further Reading · 9.1 Orthogonality

The next chapter systematically develops orthogonal matrices and orthogonal vectors

Open-Source Repository

Code and figures for the entire book: <https://github.com/Visualize-ML>

Questions Q1 – Q9

Q1 What is the essential difference between a linear transformation and an affine transformation?

Q3 A planar vector x is translated by $[1,1]$. How is this done with matrix multiplication?

Q5 Using matrix multiplication in the plane: 1) rotate 90 degrees counterclockwise about the origin; 2) then translate by $[1,1]$. How?

Q7 Using matrix multiplication in 3D space: 1) translate by $[1,1,1]$; 2) then rotate 90 degrees counterclockwise about the x_1 axis. How?

Q9 Write the 3×3 matrix that scales the unit square by $s=2$ and then translates it by $[1,-1]$; compute the transformed coordinates of its 4 vertices.

Q2 Why is translation not a linear transformation?

Q4 To undo the translation in Q3, how do you use vector subtraction and matrix multiplication respectively?

Q6 Using matrix multiplication in the plane: 1) translate by $[1,1]$; 2) then rotate 90 degrees counterclockwise about the origin. How?

Q8 Rewrite a 2×2 rotation matrix and a 2×2 shear matrix in 3×3 homogeneous-coordinate form, and verify the determinant is unchanged.

Questions Q10 – Q17

Q10 Build a 3×3 homogeneous translation matrix with numpy, and verify that its inverse simply replaces t with $-t$.

Q12 Use block matrix multiplication to prove that $\begin{bmatrix} I & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}$ equals $\begin{bmatrix} x+t \\ 1 \end{bmatrix}$.

Q14 Use code to compare: rotate 90 degrees then translate by $[1,0]$, versus translate by $[1,0]$ then rotate 90 degrees — is the final matrix the same?

Q16 Use matplotlib to visualize a triangle passing through scaling, rotation, and translation in sequence as one affine transform.

Q11 Write a Python function `homogeneous(x)` that converts a 2D column vector into homogeneous-coordinate form.

Q13 Write a Python function `affine_matrix(A, t)` that takes a 2×2 matrix A and translation vector t and returns the corresponding 3×3 affine matrix.

Q15 Build a 4×4 homogeneous matrix for a 3D composite: rotate 45 degrees about the x_3 axis, then translate by $[0,0,1]$.

Q17 Write a Python function `is_affine_invertible(A)` that checks whether a given 3×3 affine matrix is invertible.

Questions Q18 – Q25

Q18 Explore: put the 3×3 matrices for all 11 planar transforms in Table 1 into a Python list, and batch-compute their determinants.

Q20 Write a Python function `compose_affine(*matrices)` that composes multiple affine transform matrices in sequence.

Q22 Use code to demonstrate: in 3D, shearing along the x_1 axis then translating versus reversing the order — is the final position the same?

Q24 Explore affine transforms in image processing: use Python to apply a rotation-plus-translation affine transform to an image and display the result.

Q19 Use code to verify: for orthogonal projection and mirror matrices in 3×3 homogeneous form, the top-right translation entries are always 0.

Q21 Explore: given an affine matrix, how do you separate out its linear part A and translation part t ?

Q23 Comprehensive exercise: write a Python function `transform_polygon(points, M)` that applies an affine transform to a polygon's vertices in batch using homogeneous coordinates.

Q25 Preview: the next chapter introduces orthogonality (9.1) — how do this section's rotation and projection matrices relate to orthogonal matrices?

10 Prompts to Explore with AI

- 1 Explain the essential difference between linear and affine transformations, and why translation is not a linear transformation.
- 2 Explain the principle of homogeneous coordinates — why does lifting an n -D vector to $n+1$ dimensions turn translation into matrix multiplication?
- 3 Use Python to demonstrate how a 3×3 matrix can simultaneously perform planar scaling, rotation, and translation as a composite transform.
- 4 Explain why matrix multiplication generally does not commute — how do translate-then-rotate and rotate-then-translate differ?
- 5 Explain how the block matrix $\begin{bmatrix} I, t; 0, 1 \end{bmatrix}$ recovers the intuitive vector-addition equation $x+t=z$.

10 Prompts to Explore with AI

6 Explain how the 4×4 matrix for 3D affine transforms extends the 3×3 matrix for linear transforms.

7 Compare the determinants of all 11 planar transforms in Table 1, and explain the geometric meaning of their sign and magnitude.

8 Explain why projection-type transforms always have determinant 0, and whether such transforms are invertible.

9 Explain typical applications of affine transformations in computer graphics and image processing.

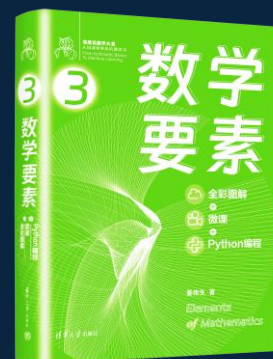
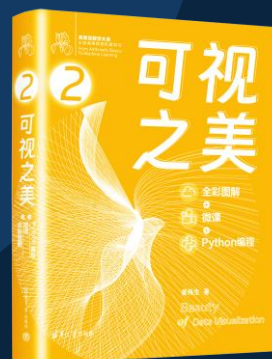
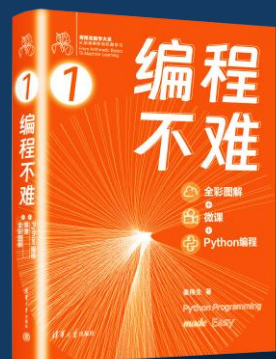
10 Use Python to visualize the same square passing through multiple affine transforms from Table 1, compared side by side.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

